Brain MRI Image Segmentation Using K-Means Clustering

Intensity-Based Unsupervised Learning, Model Selection, and Cross-Image Transfer

RESEARCH PORTFOLIO REPORT

Keman Xiang

kemanxiang005@gmail.com

June 2023

Business Analytics / Data Analytics

Computational environment: R · RStudio · R Markdown

Data: MIT 15.071 The Analytics Edge, Unit 6 recitation image matrices

Evidence note: Empirical model results are transcribed from the executed R Markdown outputs

Abstract

This study evaluates **intensity-only K-Means segmentation** of two grayscale **brain MRI** image matrices and tests the transfer of a fixed **five-centroid** representation across images.

The **healthy-image matrix** contains 365,636 pixels and the **tumor-labeled matrix** contains 292,352. Model selection favors **five clusters** as an interpretable reference rather than a statistically unique optimum: at k = 5, healthy-image WCSS is 302.1291 and explained intensity variance is **94.7683%**. A ten-seed single-start sensitivity check shows relative WCSS ranges of 0.6230% (k = 4) and 0.3112% (k = 5). Transferring the five healthy-image centroids without refitting reveals a **25.245-percentage-point decline** in the second-lowest intensity cluster and increases per-pixel reconstruction RMSE from 0.028746 to 0.039661 (**38.0%**). These findings establish measurable changes in grayscale composition and codebook fit, not anatomical segmentation accuracy or tumor detection.

The case illustrates **R analytics**, **parsimonious unsupervised model selection**, **transfer monitoring**, and **disciplined interpretation** under dataset shift.

Keywords: unsupervised learning; K-Means; image segmentation; MRI; model selection; R; data analytics; transferability

Table of Contents

Abstract 2

Table of Contents 3

1. Introduction and Research Motivation 4

    Research questions 4

2. Data, Provenance, and Unit of Analysis 4

3. Methodology 5

    3.1 Representation and computational rationale 5

    3.2 K-Means estimation and diagnostics 6

    3.3 Cross-image assignment protocol 6

4. Empirical Findings 6

    4.1 Data exploration 6

    4.2 Model-selection trade-off 6

    4.3 Learned intensity codebook and segment composition 7

    4.4 Quantization error under fixed-centroid transfer 8

    4.5 Tail residuals and zero-background sensitivity 8

    4.6 Initialization sensitivity and model-selection robustness 9

5. Discussion 9

    5.1 What the results demonstrate 9

    5.2 Relevance to business and data analytics 9

    5.3 Limitations and validity boundaries 10

6. Conclusion 10

Data, Code, and Reproducibility Statement 10

References 11

1. Introduction and Research Motivation

Medical images are high-dimensional measurements in which visual patterns must be translated into structured representations before systematic comparison. **Image segmentation** is one way to partition pixels into groups with shared characteristics. A deliberately simple **intensity-based model** offers a useful methodological case: it is interpretable, scalable relative to full pairwise hierarchical clustering, and exposes the distinction between a numerical grouping and a meaningful clinical label [1, 2].

Our central question is not whether K-Means diagnoses a tumor. Rather, we ask **what an unsupervised model can reliably establish from grayscale intensities**, **how the number of groups changes representation quality**, and **how a fixed representation behaves when moved to a second image**. These questions parallel common business-analytics challenges in segmentation, quantization, unsupervised customer grouping, and deployment under distribution shift.

Research questions

RQ1. What structural and distributional characteristics of the two supplied grayscale images affect pixel-level clustering?

RQ2. How do cluster compactness and intensity variance explained change as the number of K-Means clusters increases?

RQ3. What can fixed-centroid assignment reveal about a second image, and what cannot be inferred without labels or matched imaging conditions?

2. Data, Provenance, and Unit of Analysis

The study uses two CSV matrices: **healthy.csv** (566 rows × 646 columns) and **tumor.csv** (571 rows × 512 columns). Each entry is a grayscale intensity in [0, 1]. The datasets contain neither pixel-level tissue masks nor imaging metadata, patient identifiers, registration transforms, or verified outcome labels. We use “healthy” and “tumor” as dataset identifiers, not as independent clinical diagnoses. All pixel counts and descriptive statistics in this section are calculated directly from the supplied CSV files.

Image	Dimensions	Pixels	Mean	SD	Zeros	Distinct values
Healthy	566 × 646	365,636	0.1302	0.1257	9.30%	206
Tumor-labeled	571 × 512	292,352	0.1595	0.1715	34.67%	254

Both matrices contain finite intensities and no missing values. The unequal image dimensions and sharply different proportions of zero-valued pixels matter: a raw difference in cluster counts would conflate image size with image composition. We therefore prioritize **within-image percentages** for comparisons, and we do not assume pixel-to-pixel anatomical correspondence.

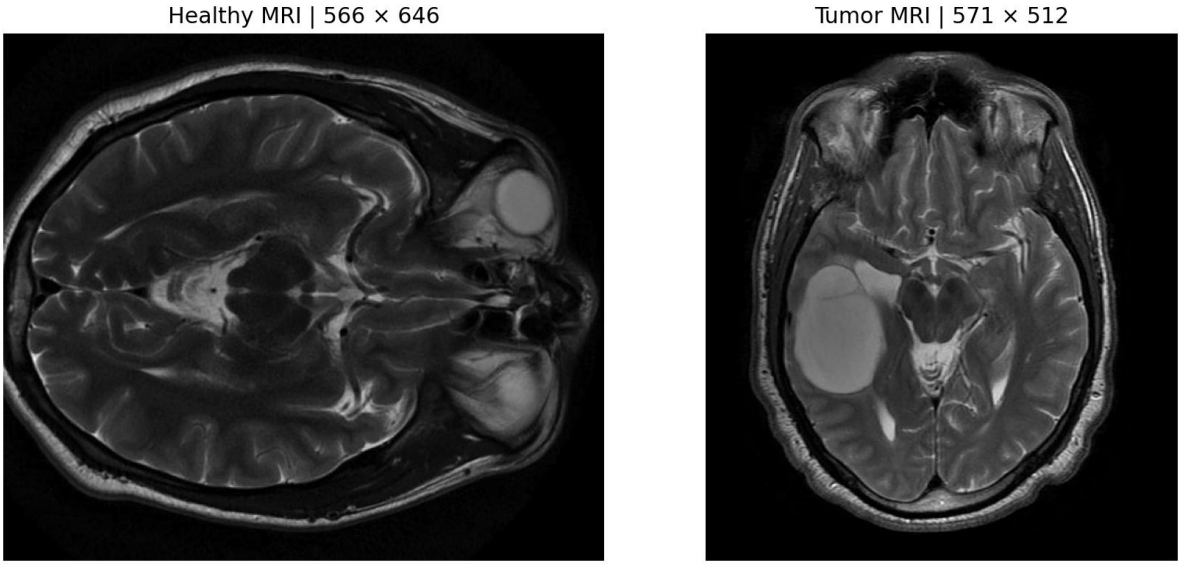


Figure 1. Original images reconstructed directly from the uploaded CSV matrices.

Display contrast is fixed to the common 0–1 intensity range.

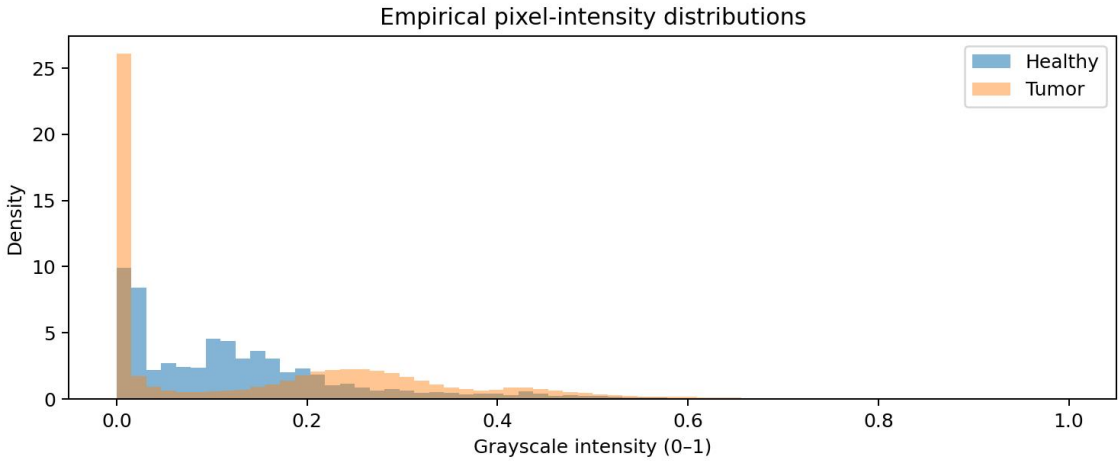


Figure 2. Observed intensity distributions.

The tumor-labeled image has a larger zero-intensity fraction than the healthy image. This difference can materially affect cluster occupancy, the apparent prevalence of darker regions, and any pooled comparison. We cannot attribute this distributional shift to a tumor because background coverage, acquisition and preprocessing may also differ.

3. Methodology

3.1 Representation and computational rationale

We represent each image as a **two-dimensional matrix** and **vectorize** it into **one grayscale value per pixel**, preserving the original matrix for spatial reconstruction. In R, as.vector() uses column-major ordering; the same dimensions must be used to restore assignments to the image. The feature space is one-dimensional: spatial coordinates, neighborhood consistency, texture, and anatomical landmarks are not included.

A full Euclidean pairwise-distance representation for 365,636 pixels requires approximately 66.8 billion distinct distances (roughly 498 GiB at eight bytes per distance, excluding overhead). We therefore use **K-Means** rather than attempting to materialize a full hierarchical distance object for the healthy MRI.

3.2 K-Means estimation and diagnostics

For a specified **k**, K-Means assigns each intensity x_i to a centroid μ_j to minimize $WCSS = \Sigma_j \Sigma_i \in C_j (x_i - \mu_j)^2$. The five-cluster reference uses set.seed(1), kmeans(healthy_vector, centers = 5, iter.max = 1000), while the model-selection sweep evaluates **k = 2 through 10** using five random starts and the Hartigan–Wong algorithm. We evaluate WCSS, the proportion of total intensity sum of squares explained by between-cluster variation, marginal WCSS reductions, convergence status, visual segmentations, and initialization sensitivity.

We interpret "explained variance" as in-sample pixel-intensity quantization, not prediction accuracy or medically meaningful variance. The elbow heuristic cannot by itself identify a uniquely optimal k: more clusters normally improve the fit even when they add limited interpretability [3].

3.3 Cross-image assignment protocol

We estimate **five centroids** on the **healthy** image only. After ordering the centroids by brightness, we assign each pixel in the **tumor**-labeled image to its nearest fixed centroid, using the midpoint between adjacent ordered centroids as a one-dimensional decision boundary. We do not refit the centroids on the second image. We compare cluster occupancy using within-image proportions, and assess quantization fit using per-pixel MAE, MSE, and RMSE. These metrics quantify representation error, not segmentation overlap or clinical accuracy.

4. Empirical Findings

4.1 Data exploration

The **healthy** matrix contains 365,636 observed pixels and the **tumor**-labeled matrix contains 292,352. Their means are 0.1302 and 0.1595, respectively; their sample standard deviations are 0.1257 and 0.1715. The zero-intensity shares are 9.30% and 34.67%. These are descriptive properties of two image matrices, not statistical estimates across patients.

4.2 Model-selection trade-off

k	WCSS	Intensity variance explained	WCSS reduction vs. prior k	Convergence code
4	467.8845	91.8980%	44.5445%	0
5	302.1291	94.7683%	35.4266%	0
6	220.7184	96.1780%	26.9457%	0



Figure 3. R model-selection results for k = 4, 5, and 6. These are a subset of the k = 2–10 sweep.

The **44.54%** reduction at **k = 4** refers to the improvement from k = 3 to k = 4; it is not a direct comparison between k = 4 and k = 5. Adding a fifth cluster lowers WCSS by **35.43%** relative to k = 4, while adding a sixth lowers it by 26.95% relative to k = 5. The **gain** in explained intensity variance is **2.87** percentage points **from four to five clusters** and 1.41 points from five to six. All three models report convergence code 0.

These findings support **k = 4** and **k = 5** as credible parsimonious candidates. We retain **k = 5** as the cross-image reference because it preserves more intensity detail than k = 4 at a moderate additional complexity cost; however, the still-material k = 5-to-6 WCSS reduction means the reported values do not prove a sharp elbow or unique optimum at five. A final choice also depends on the full k = 2–10 curve, image interpretability, and initialization stability. The ten-seed single-start sensitivity analysis reports relative WCSS ranges of 0.62304% for k = 4 and **0.31117%** for k = 5. These small but nonzero ranges support reporting multiple starts rather than assuming that every initialization returns an identical optimum.

4.3 Learned intensity codebook and segment composition

The fitted healthy-image model yields **five ordered grayscale centroids** (darkest to brightest). Cluster numbering in the table is by **intensity rank**, not the arbitrary identifiers returned by K-Means. Each share uses the corresponding image's entire pixel count as its denominator, including zero-valued background.

Intensity cluster	Healthy centroid	Healthy pixels	Tumor pixels	Healthy share	Tumor share	Difference (pp)
1 (darkest)	0.02189	139,963	135,357	38.279%	46.299%	+8.020
2	0.11656	114,480	17,731	**31.310%**	**6.065%**	**−25.245**
3	0.20136	65,557	58,506	17.930%	20.012%	+2.083
4	0.32991	27,338	52,220	7.477%	17.862%	+10.385
5 (brightest)	0.49139	18,298	28,538	5.004%	9.762%	+4.757

The largest compositional change occurs in **cluster 2**, which falls from 31.310% to 6.065% (**−25.245 percentage points**). The fourth intensity cluster increases by 10.385 points, and the darkest cluster increases by 8.020 points. These shifts are measured against the same five healthy-image centroids; they therefore

indicate changes in the distribution of assigned grayscale categories rather than changes caused by independently refitting a model. The sum of share differences is approximately zero, subject to displayed rounding. No cluster can be interpreted as a verified tissue type or tumor region without clinical re-examination.

4.4 Quantization error under fixed-centroid transfer

We compare each observed **pixel intensity** with its assigned healthy-image centroid. The error metrics describe grayscale reconstruction fidelity; they do not compare a predicted mask against a clinical ground-truth segmentation.

Image	MAE	MSE	RMSE	Total WCSS
Healthy (training)	0.022047	0.000826	**0.028746**	302.1291
Tumor (transfer)	0.029319	0.001573	**0.039661**	459.8580

Per-pixel RMSE increases by approximately **38.0%** (0.039661 versus 0.028746), while MAE increases by approximately 33.0%. MSE rises by approximately 90.4%, reflecting the quadratic penalty on larger residuals. The transfer image also has fewer pixels than the training image, so total WCSS is not a size-normalized comparison; MAE, MSE, and RMSE are more appropriate for comparing the two matrices. Higher transfer error shows that the original grayscale codebook approximates the second image less closely; it does not establish pathology, a failure of clinical segmentation, or a patient-level generalization estimate.

4.5 Tail residuals and zero-background sensitivity

Image	Median absolute error	90th-percentile absolute error	Pixels beyond centroid range
Healthy	0.02085	0.04223	25.6356%
Tumor	0.02189	0.05540	46.4382%

Although median absolute errors differ only modestly, the 90th-percentile residual is larger on the transfer image (0.05540 versus 0.04223). The fraction of pixels outside the observed range of the five healthy-image centroids rises from 25.6356% to 46.4382%. This supports the interpretation that tail intensities are less well represented in the second image, but the beyond-range statistic should not be conflated with an out-of-distribution patient diagnosis.

Image	Exact-zero share	Positive-intensity pixels	Positive-only mean	Positive-only median
Healthy	9.29504%	331,650	0.14358	0.11966
Tumor	34.66917%	190,996	0.24417	0.24314

The second image contains 34.66917% exact-zero pixels, versus 9.29504% in the first—a 25.374-percentage-point difference. Removing exact zeros descriptively increases the observed mean intensity to 0.14358 and 0.24417, respectively. This sensitivity check reveals that both background prevalence and positive-intensity composition differ. However, nonzero pixels are not a validated brain-only mask, and these image-wide contrasts cannot be attributed causally to tumor status.

4.6 Initialization sensitivity and model-selection robustness

k	Single-start runs	Minimum WCSS	Median WCSS	Maximum WCSS	Relative range
4	10	467.8845	470.7996	470.7996	0.62304%
5	10	301.7859	302.7250	302.7250	0.31117%

The minimum single-start WCSS at k = 5 (301.7859) differs slightly from the k = 5 reference-model WCSS (302.1291), as the experiments use different initialization protocols. We retain each reported result with its own experimental context rather than silently replacing the fixed reference centroids. A low spread across these ten seeds indicates modest observed sensitivity in this specific image, not proof of global optimality or external validity.

5. Discussion

5.1 What the results demonstrate

The study demonstrates a complete **unsupervised analytics workflow:** validate raw measurements; select a tractable representation; compare competing levels of granularity; distinguish fit from substantive validity; and carry a fixed representation to a new sample.

K-Means turns hundreds of thousands of grayscale measurements into a small set of interpretable intensity levels. **Model-selection** diagnostics show the benefit of additional centroids while also revealing diminishing marginal gains. **The cross-image protocol** tests representation transfer without leaking target-image observations into training.

In the executed results, **cluster 2** drops by 25.245 percentage points and the transfer RMSE rises by 38.0%, demonstrating measurable codebook mismatch, indicating, though does not prove, a clinical abnormality detection.

5.2 Relevance to business and data analytics

The substantive domain is **medical imaging**, but the analytic decisions are general. Cluster-count selection resembles choosing **customer-segmentation granularity:** a finer partition fits observed heterogeneity more closely but may reduce parsimony and practical usability. Fixed-centroid transfer resembles assigning new customers, products, or transactions to an established segmentation scheme, allowing changes in segment composition to be monitored without redefining the categories at each measurement. The observed 25.245-point cluster-share shift illustrates why business segmentation monitoring should distinguish changes in customer or product composition from shifts in data collection, scaling, or population mix. The 38.0% increase in reconstruction RMSE offers a complementary model-monitoring signal, but is not a task-outcome metric.

The **project** therefore provides evidence of **R-based statistical programming**, **large-vector data handling**, **diagnostic evaluation**, and **critical interpretation**. Though it does not establish clinical expertise, validated medical decision support, or causal effects, its value for a business-analytics research portfolio lies in the transparency of the analytical choices and the restraint of the conclusions.

5.3 Limitations and validity boundaries

First, two images are not a patient cohort; treating hundreds of thousands of pixels as independent patients would constitute pseudoreplication.

Second, grayscale alone cannot enforce spatial continuity or separate tissues with overlapping intensities.

Third, the images differ in size, background composition, and potentially acquisition conditions; without registration and metadata, apparent changes cannot be assigned to pathology.

Fourth, no ground-truth masks exist, so Dice, intersection-over-union, sensitivity, specificity, and clinical tumor-detection accuracy cannot be estimated [4].

Fifth, K-Means is sensitive to initialization and k; WCSS rewards additional clusters by construction.

Finally, using a healthy-image intensity codebook on a different image is a transfer of representation, not proof of generalization across scanners, patients, or clinical settings.

6. Conclusion

This project shows that a transparent, intensity-only **K-Means model** can compress hundreds of thousands of **grayscale pixel measurements** into **five reusable numerical categories**. The reference model achieves **WCSS 302.1291** and explains **94.7683%** of healthy-image intensity variation, while comparisons with k = 4 and k = 6 demonstrate a meaningful but non-unique fit–complexity trade-off. Ten-seed sensitivity checks show relative WCSS ranges below 0.7% for both k = 4 and k = 5 in the tested configuration.

The principal empirical contribution is the **fixed-centroid cross-image comparison**. Under a common **five-centroid** codebook, **cluster 2** decreases by **25.245 percentage** points and cluster 4 increases by 10.385 points. The second image has higher reconstruction error (RMSE 0.039661 versus 0.028746) and a higher 90th-percentile absolute residual (0.05540 versus 0.04223). At the same time, exact-zero pixels account for 34.669% of the second image versus 9.295% of the first, highlighting background and preprocessing as potential confounders. These numerical results may support an inference about the location, presence, or biological properties of a **tumor**, but it is not definite without clinical evidence afterwards.

For business and data analytics, the transferable lesson is methodological: fit an interpretable segmentation on a reference sample, choose its complexity using several diagnostics, preserve the segment definitions during transfer, and evaluate both composition drift and representation error.

For medical imaging, the next stage would require multiple independently sourced cases, harmonized acquisition, anatomical masks, spatially aware features, and clinically annotated ground truth before diagnostic claims or validated segmentation metrics could be considered.

Data, Code, and Reproducibility Statement

Source inputs:

- *healthy.csv;*
- *tumor.csv;*
- *Unit6_Recitation.pdf;*
- *Unit6_Recitation.R;*
- *week6_3.2.R.*

R Analysis artifacts:

- 01_Data_Exploration.Rmd;

- 02_KMeans_Segmentation.Rmd;

- 03_Model_Selection.Rmd;

- 04_Cross_Image_Analysis.Rmd.

*We use **RStudio** and **R Markdown** for executable analysis and **HTML** output.*

Report tables derived from the CSVs are independently computed from those same files; all clustering, model-selection, cross-image occupancy, residual, and initialization tables reproduce the executed outputs of the four R Markdown HTML/PDF reports. R 4.2.1 and the recorded R Markdown session information are documented in the executed analysis. We do not substitute outside MRI data.

This report is a computational portfolio case study, not a clinical validation study.

References

[1] MIT OpenCourseWare. 15.071 The Analytics Edge (Spring 2017), Unit 6 Recitation: Seeing the Big Picture—Segmenting Images to Create Data. Course PDF, R script, and supplied grayscale CSV matrices. https://ocw.mit.edu/courses/15-071-the-analytics-edge-spring-2017/

[2] Nature Methods. Content types and guidelines for Analysis articles. https://www.nature.com/nmeth/content

[3] Google for Developers. Machine Learning: Clustering—Evaluating Results. https://developers.google.com/machine-learning/clustering/kmeans/evaluate-results

[4] Reinke, A. et al. (2022). Towards a guideline for evaluation metrics in medical image segmentation. BMC Research Notes. https://doi.org/10.1186/s13104-022-06096-y

[5] Springer Nature. Structuring your manuscript: IMRaD. https://www.springernature.com/gp/authors/campaigns/writing-a-manuscript/structuring-your-manuscript

[6] Australian National University. Science reports: Methods, Results, Discussion. https://www.anu.edu.au/students/academic-skills/writing-and-assessment/science-reports

[7] Maier-Hein, L. et al. Metrics reloaded: recommendations for image analysis validation. Nature Methods 21, 195–212. https://doi.org/10.1038/s41592-023-02151-z

[8] Reinke, A. et al. Understanding metric-related pitfalls in image analysis validation. Nature Methods 21, 182–194. https://doi.org/10.1038/s41592-023-02150-0